

NovaCart

India's Intelligent Online Marketplace

Smart Shopping. Smarter Decisions.

Business Requirements Document

Profit Recovery Analytics Platform

BRD Detail	Value
Version	1.0
Period	FY2025-FY2026
Prepared by	Aishi Dutta
Purpose	Master blueprint for SQL, Python, Machine Learning and Power BI delivery

Source of truth: 01_Company_Profile | Headquarters: Bengaluru, Karnataka, India

Document Control

Control field	Detail
Document title	Business Requirements Document
Project	Profit Recovery Analytics Platform
Company	NovaCart - India's Intelligent Online Marketplace
Source document	01_Company_Profile.docx / 01_Company_Profile.pdf
Business owner	Executive Management
Delivery owner	Data and Analytics
Primary audience	Executive leadership, business stakeholders, recruiters and BI portfolio reviewers
Status	Version 1.0 portfolio baseline
Disclaimer	Fictional case-study artifact for educational and portfolio use

Important disclaimer. This document describes a fictional company created solely for educational and portfolio purposes. All names, figures, processes, business metrics and operational details are synthetic and do not represent any real organization.

Version History

Version	Period	Change	Description	Prepared by
1.0	FY2026	Initial BRD baseline	Defines scope, requirements, data, reporting and acceptance criteria for the BI project.	Aishi Dutta

Table of Contents

Cover Page.....1

Document Control.....2

Version History.....2

Table of Contents.....3

Executive Summary.....4

Business Background.....5

Current Business Situation.....5

Problem Statement.....6

Business Objectives.....6

Project Objectives.....7

Project Scope.....7

Out of Scope.....8

Assumptions.....8

Constraints.....8

Risks.....9

Stakeholder Overview.....10

Business Drivers.....11

Expected Benefits.....11

Functional Requirements.....12

Non-Functional Requirements.....13

Data Requirements.....13

Business Rules.....14

Reporting Requirements.....14

Analytics Requirements.....15

Dashboard Requirements.....15

Success Metrics.....16

Deliverables.....16

Acceptance Criteria.....17

Project Timeline.....17

Appendix.....18

Executive Summary

This BRD defines the business and analytics requirements for NovaCart's Profit Recovery Analytics Platform. The project exists because revenue, orders and customer growth have increased while profit, delivery speed, returns performance and customer satisfaction have moved in the wrong direction.

The document translates the Company Profile into an executable BI blueprint. It defines the scope, stakeholders, requirements, data assets, business rules, dashboards, success metrics and acceptance criteria needed for later SQL, Python, Machine Learning exploration and Power BI work.

BI Analytics Blueprint

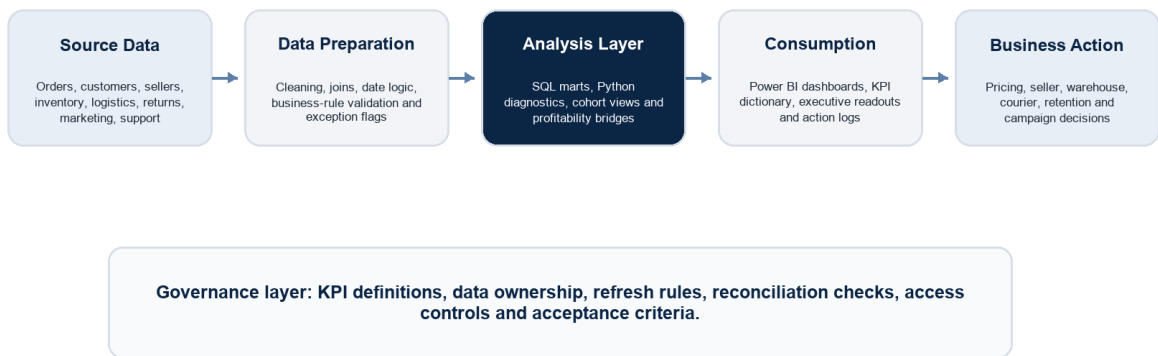


Figure 1. Analytics blueprint showing how source data becomes decisions.

Consulting position. The BRD should be treated as the control document for all downstream analysis. No metric, dashboard page or model feature should be introduced unless it can be traced back to a business question, rule or KPI in this document.

Business Background

NovaCart is a fictional B2C multi-seller marketplace headquartered in Bengaluru, Karnataka, India. The company operates through website, android and ios and serves demand through 520 active sellers, 24,500 products and 8 warehouses.

Business area	Established profile fact
Customers	185,000 registered customers generating browse, order, review, support and return signals.
Marketplace	520 active sellers across 12 product categories.
Operations	8 warehouses process allocation, pick-pack, dispatch, delivery and return flows.
Financial context	FY2025 revenue is ₹185 Crore and profit is ₹16.5 Crore.

Current Business Situation

The situation is a growth-quality problem, not a simple demand problem. The profile signals indicate expansion in volume alongside deterioration in profit quality and customer experience.

Business signal	Year-over-year change
Revenue	Increased by 31%
Orders	Increased by 28%
Customers	Increased by 22%
Profit	Decreased by 18%
Marketing spend	Increased by 42%
Product returns	Increased by 29%
Average delivery time	Increased from 2.9 to 4.8 days
Customer satisfaction	Decreased from 4.6 to 3.9
Customer support tickets	Increased by 36%

Problem Statement

Problem statement. NovaCart does not yet have an integrated analytical view explaining why profit has declined despite higher revenue, orders and customers. Leadership needs a reliable BI foundation that connects sales, marketing, logistics, returns, inventory, seller performance and customer experience into one decision system.

Without this blueprint, later dashboards risk becoming isolated visuals rather than decision tools. The BRD therefore defines what must be measured, how it should be governed and how outputs will be accepted by stakeholders.

Business Objectives

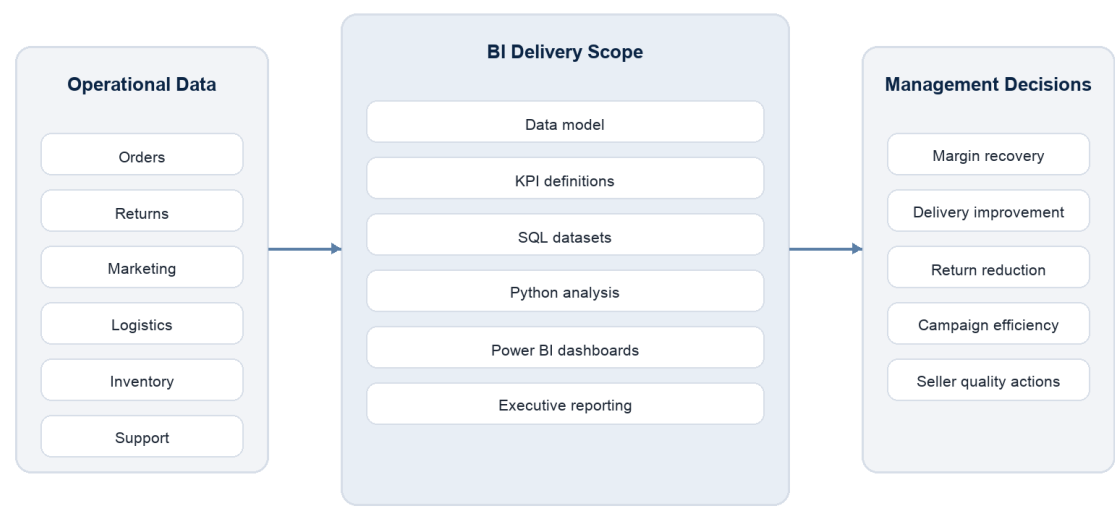
KPI	Current	Target	Strategic rationale	Owner	Measurement method
Revenue Growth	31%	40%	Scale profitable demand	Executive / Marketing	Monthly revenue dashboard
Profit Margin	9%	14%	Recover contribution quality	Finance	P&L and margin bridge
Return Rate	12%	Below 8%	Reduce reverse logistics cost	Seller / Operations	Return reason analytics
Average Delivery Time	4.8 days	3 days	Improve experience and repeat orders	Logistics	Courier SLA dashboard
Customer Satisfaction	3.9/5	4.6/5	Restore customer trust	Support	Post-delivery CSAT
Repeat Purchase Rate	34%	50%	Grow retention economics	Marketing	Cohort repeat analysis
Marketing ROI	2.4x	3.5x	Shift spend to efficient campaigns	Marketing	Campaign ROI model
Customer Acquisition Cost	₹1,280	₹950	Lower paid acquisition burden	Marketing / Finance	CAC waterfall report

Project Objectives

ID	Project objective	Downstream use
PO-01	Create a governed KPI and data foundation for profit recovery decisions.	Executive reporting, KPI dictionary, Power BI
PO-02	Identify where profit leakage may occur across categories, returns, logistics, marketing and seller quality.	SQL marts, Python diagnostics
PO-03	Translate analysis into measurable recommendations aligned to FY2026 targets.	Executive readout, action tracking
PO-04	Prepare clean, reusable datasets for future machine learning exploration.	Feature-ready analytical tables

Project Scope

BRD Scope Boundary



Excluded: live platform changes, payment gateway engineering, partner-specific performance publication and production ML deployment.

Figure 2. Scope boundary for the BI project.

Scope element	Definition
In scope	Profitability, sales, marketing ROI, seller performance, warehouse efficiency, delivery performance, returns and refunds, customer retention, inventory optimization and executive reporting.
Core tools	SQL, Python, Excel, Power BI and structured documentation.
Primary outputs	Requirements, KPI dictionary, data model, analysis datasets, dashboards, insights and recommendations.

Out of Scope

Excluded area	Reason
Platform engineering	Changes to website, Android, iOS, order management, warehouse management or payment systems.
Partner operations	Courier contract negotiation or partner-specific public performance scorecards.
Production ML deployment	The BRD supports future model exploration but does not define a production machine learning release.
Financial audit	The project analyses synthetic business performance and does not certify accounting records.

Assumptions

- The Company Profile remains the approved source for NovaCart facts, metrics, departments and processes.
- Required operational data can be represented through synthetic portfolio datasets for analysis and dashboarding.
- Stakeholders will validate KPI definitions before dashboard build begins.
- Data quality exceptions will be documented rather than silently corrected inside visuals.

Constraints

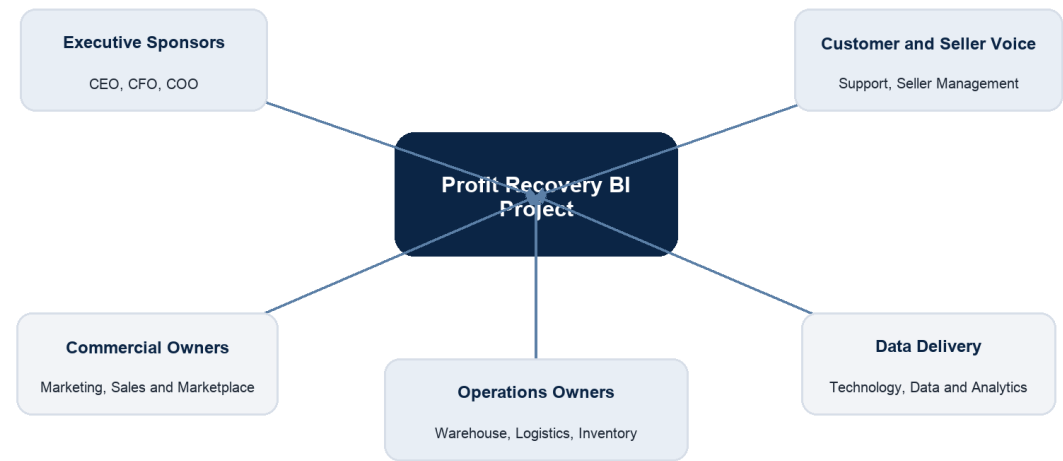
Constraint	BRD implication
Synthetic portfolio context	All figures and records must remain fictional and internally consistent.
No real employee names	Stakeholder ownership is defined by role or department only.
Tooling scope	Later work should use SQL, Python, Machine Learning exploration and Power BI deliverables.
Executive readability	Outputs must prioritize decision clarity over excessive technical detail.

Risks

ID	Risk	Impact	Rating	Mitigation	Owner
R-01	Metric ambiguity	Different teams may define profit, returns or ROI differently.	High	Approve KPI dictionary before dashboard build.	Data and Analytics
R-02	Data quality gaps	Missing joins or inconsistent timestamps may distort analysis.	High	Profile source data and document exclusions.	Technology
R-03	Scope expansion	Stakeholders may request operational system changes inside a BI project.	Medium	Use out-of-scope controls and change log.	Executive Management
R-04	Attribution error	Marketing or logistics changes may be over-credited for profit movement.	Medium	Separate correlation, diagnostic evidence and recommendations.	Data and Analytics
R-05	Low adoption	Dashboards may not change decisions if owners are unclear.	Medium	Assign KPI owners and review cadence.	Executive Management
R-06	Privacy exposure	Customer data may be shown at unnecessary granularity.	Medium	Use masking, aggregation and role-based access.	Technology
R-07	Overbuilt models	ML exploration may distract from basic profit recovery diagnostics.	Low	Sequence descriptive and diagnostic analysis first.	Data and Analytics

Risk posture. The most important BRD control is traceability: every dashboard, dataset and analytical conclusion must map back to an approved requirement or business question.

Stakeholder Engagement Model



Engagement rhythm: executive steering, SME validation, data-quality review and adoption feedback.

Figure 3. Stakeholder engagement model for the BRD.

Stakeholder group	Role	Primary responsibility	Engagement cadence
Executive Management	Sponsor	Approve direction, targets and action ownership	Monthly steering
Finance	Business owner	Validate profit margin, cost and settlement logic	Weekly during definition
Operations	SME	Validate warehouse and fulfilment performance logic	Weekly during discovery
Logistics	SME	Validate delivery, RTO, damage and SLA metrics	Weekly during discovery
Marketing	SME	Validate spend, CAC, ROI and retention logic	Weekly during analysis
Seller Management	SME	Validate seller quality and commission performance	As required
Technology	Data custodian	Confirm source availability, access and refresh feasibility	Weekly during build
Data and Analytics	Delivery owner	Build datasets, analysis, dashboards and executive outputs	Ongoing

Business Drivers

Driver	Evidence from profile	BI implication
Profit pressure	Profit decreased while volume metrics increased.	Requires margin bridge and cost-driver analysis.
Return growth	Product returns increased by 29%.	Requires return reason, seller, category and fulfilment analysis.
Delivery deterioration	Average delivery time increased from 2.9 to 4.8 days.	Requires warehouse, courier and region-level SLA diagnostics.
Marketing efficiency	Marketing spend increased by 42%.	Requires CAC, ROI, cohort and campaign-payback visibility.
Customer experience	Customer satisfaction decreased from 4.6 to 3.9.	Requires support, review, delivery and return signal integration.

Expected Benefits

Benefit	Expected value
Executive clarity	Single view of growth, profit, cost, returns and service health.
Faster diagnosis	Reusable SQL and Python datasets reduce ad hoc analysis effort.
Better accountability	Each KPI has an owner, measurement method and action path.
Portfolio depth	The BRD creates a credible foundation for later SQL, Python, ML and Power BI artifacts.
Decision discipline	Insights are tied to defined objectives rather than isolated dashboard visuals.

Functional Requirements

ID	Requirement	Owner	Downstream artifact
FR-01	Create an integrated order-level analytical dataset linking customer, seller, product, warehouse, delivery, payment and return events.	Data and Analytics	SQL data mart
FR-02	Calculate revenue, profit, return rate, delivery time, customer satisfaction, repeat purchase rate, marketing ROI and CAC using approved definitions.	Finance	KPI dictionary
FR-03	Segment performance by product category, warehouse, seller group, customer cohort, marketing channel and delivery status.	Executive Management	Power BI filters
FR-04	Identify transactions with potential profit leakage from discounting, returns, delivery delay, refund adjustment or seller quality issues.	Finance	Python diagnostics
FR-05	Provide return analysis by reason, category, seller, warehouse, fulfilment stage and settlement impact.	Operations	Returns dashboard
FR-06	Provide marketing analysis by spend, acquisition source, order conversion, repeat purchase and ROI.	Marketing	Marketing dashboard
FR-07	Provide logistics analysis for delivery time, on-time rate, first-attempt rate, damage, RTO and shipping cost per order.	Logistics	Logistics dashboard
FR-08	Provide seller performance analysis covering rating, return rate, dispatch time, cancellation rate, defect rate and commission contribution.	Seller Management	Seller dashboard
FR-09	Generate executive summaries with risks, KPI movement and recommended management actions.	Executive Management	Executive report
FR-10	Prepare feature-ready datasets for future machine learning exploration, without defining production deployment.	Data and Analytics	ML exploration table

Functional traceability controls

Control	Purpose	Implementation note
Requirement ID	Keeps business questions traceable through SQL, Python and Power BI.	Use FR IDs in notebook headings, SQL comments and dashboard documentation.
Business owner	Prevents analytics outputs from becoming ownerless artifacts.	Each requirement has a validating department or executive stakeholder.
Downstream artifact	Clarifies where each requirement will appear in the portfolio.	Avoid building visuals or models that do not map to a defined requirement.

Non-Functional Requirements

ID	Requirement area	Requirement statement
NFR-01	Usability	Executive dashboards must be readable without technical explanation and support drill-down for analysts.
NFR-02	Performance	Dashboard pages should load quickly for common filters and avoid unnecessary high-cardinality visuals.
NFR-03	Security	Role-based access, masking and aggregation must protect customer and seller-sensitive fields.
NFR-04	Auditability	KPI logic, filters, exclusions and refresh timing must be documented.
NFR-05	Maintainability	SQL, Python and Power BI logic should use consistent naming and reusable transformations.
NFR-06	Data quality	Missing, duplicate or inconsistent records must be profiled and flagged before analysis.

Data Requirements

Data domain	Required fields	Granularity	Refresh
Orders	Order ID, customer ID, seller ID, product ID, warehouse ID, order date, amount, discount, status	Order-level	Daily
Returns	Return request, reason, eligibility, pickup, inspection, approval, refund and adjustment fields	Return-level	Daily
Products	Category, product, price band, listing quality and defect flags	Product-level	Weekly
Sellers	Rating, dispatch time, cancellation, defect rate and commission contribution	Seller-level	Weekly
Warehouses	Allocation, pick-pack time, dispatch SLA and exception status	Warehouse/order	Daily
Logistics	Pickup, transit, delivery, failed delivery, RTO, damage and cost fields	Shipment-level	Daily
Marketing	Channel, campaign, spend, acquisition source, conversion and repeat-order fields	Campaign/cohort	Weekly
Support	Ticket date, reason, severity, resolution time, refund linkage and satisfaction signal	Ticket-level	Daily

Data readiness controls

Control	Minimum check	Reason
Reconciliation	Order, product, seller and warehouse totals match approved profile values.	Protects consistency across portfolio deliverables.
Completeness	Required keys and dates are populated for analytical joins.	Prevents false drops in SQL marts.
Validity	Return reasons, status values and delivery events follow approved business rules.	Improves confidence in diagnostic analysis.
Lineage	Every dashboard metric can be traced to source fields and transformation logic.	Supports recruiter and executive review.

Business Rules

ID	Rule
BR-01	Returns are accepted within seven days for damaged product, wrong product delivered, manufacturing defect, not as described, fashion size or fit issue and delivery delay.
BR-02	Product category totals must equal 24,500 and must reconcile to the Company Profile.
BR-03	Warehouse count must remain eight and use WH001 through WH008 as defined in the Company Profile.
BR-04	Partner names may be used only illustratively and must not imply real partnerships.
BR-05	Partner-specific performance results must not be published in the portfolio deliverables.
BR-06	Profit recovery findings must distinguish confirmed evidence from investigation hypotheses.
BR-07	Customer-level fields must be masked or aggregated in executive reporting.
BR-08	Any future metric change must update the KPI dictionary and downstream dashboards.

Reporting Requirements

ID	Report	Audience	Cadence	Primary content
RR-01	Executive weekly profit recovery summary	CEO, CFO, COO	Weekly	KPI movement, risks, actions and owner status
RR-02	Monthly marketplace performance pack	Executive team	Monthly	Category, seller, customer and financial trends
RR-03	Operations exception report	Operations, warehouse, logistics	Weekly	Delivery delays, dispatch breaches, damage and RTO patterns
RR-04	Marketing efficiency report	Marketing and finance	Monthly	Spend, CAC, ROI, cohort and repeat purchase view
RR-05	Returns and refund leakage report	Finance, support, seller management	Weekly	Return reason, refund, inspection and settlement impact

Analytics Requirements

ID	Analysis	Business question answered
AR-01	Profit bridge	Explain movement from revenue growth to profit decline using cost, return, discount and operational drivers.
AR-02	Return diagnostics	Identify categories, reasons, sellers and fulfilment stages associated with return growth.
AR-03	Delivery impact analysis	Quantify how delivery delays relate to satisfaction, support tickets, returns and repeat purchase.
AR-04	Marketing efficiency analysis	Compare channel spend, CAC, ROI and repeat-order quality.
AR-05	Seller quality analysis	Rank seller groups by return, defect, cancellation and commission contribution risk.
AR-06	Customer retention analysis	Build cohorts for first order, repeat purchase and support/return experience.
AR-07	ML exploration readiness	Create candidate features for churn risk, return propensity and delivery delay risk without production deployment.

Dashboard Requirements

Dashboard page	Core KPIs and views	Decision use
Executive Overview	Profit margin, revenue growth, orders, returns, delivery time, satisfaction, CAC and ROI	Leadership decision view
Profitability	Margin bridge, category contribution, discount and refund leakage	Finance and executive review
Returns	Return reason, category, seller, warehouse and refund outcome	Operations, support and seller actions
Logistics	Delivery time, on-time rate, first-attempt rate, damage, RTO and cost	Logistics and operations recovery
Marketing	Spend, CAC, ROI, acquisition source, repeat purchase and cohort quality	Campaign optimization
Seller Performance	Rating, return rate, dispatch, cancellation, defect and commission contribution	Seller quality governance
Customer Experience	CSAT, support tickets, repeat purchase, reviews and return/support paths	Retention improvement

Dashboard interaction standards

Standard	BRD expectation
Executive landing view	Start with KPI health, movement, risks and recommended actions before detailed drill-down.
Consistent filters	Use category, warehouse, seller group, channel, date and customer cohort filters consistently across pages.
Drill-through	Allow executives to move from summary indicators to root-cause views without leaving the report.
Evidence labelling	Separate confirmed findings, investigation areas and recommendations in dashboard text.

Success Metrics

KPI	Current	Target	Strategic rationale	Owner	Measurement method
Revenue Growth	31%	40%	Scale profitable demand	Executive / Marketing	Monthly revenue dashboard
Profit Margin	9%	14%	Recover contribution quality	Finance	P&L and margin bridge
Return Rate	12%	Below 8%	Reduce reverse logistics cost	Seller / Operations	Return reason analytics
Average Delivery Time	4.8 days	3 days	Improve experience and repeat orders	Logistics	Courier SLA dashboard
Customer Satisfaction	3.9/5	4.6/5	Restore customer trust	Support	Post-delivery CSAT
Repeat Purchase Rate	34%	50%	Grow retention economics	Marketing	Cohort repeat analysis
Marketing ROI	2.4x	3.5x	Shift spend to efficient campaigns	Marketing	Campaign ROI model
Customer Acquisition Cost	₹1,280	₹950	Lower paid acquisition burden	Marketing / Finance	CAC waterfall report

Deliverables

ID	Deliverable	Purpose
D-01	Business Requirements Document	Approved master blueprint for the BI project
D-02	Stakeholder Analysis	Role, influence, cadence and decision ownership mapping
D-03	KPI Dictionary	Metric definitions, formulas, grain, owner and source fields
D-04	SQL Scripts	Reusable marts for profitability, returns, logistics, marketing and seller analysis
D-05	Python Analysis	Notebook-based diagnostics, charts and business findings
D-06	Machine Learning Exploration	Candidate features and modelling notebook for future analytical maturity
D-07	Power BI Dashboard	Executive and functional pages aligned to this BRD
D-08	Executive Presentation	Findings, recommendations, actions and implementation priorities

Acceptance Criteria

ID	Acceptance criterion
AC-01	All company facts, KPIs, categories, departments, warehouses and processes reconcile to the Company Profile.
AC-02	Every dashboard KPI traces to a KPI definition, business rule and data source.
AC-03	SQL outputs reconcile to expected row counts, key totals and data-quality checks.
AC-04	Python analysis distinguishes evidence, hypothesis and recommendation.
AC-05	Power BI pages answer executive questions without requiring raw-data inspection.
AC-06	ML exploration uses documented features and avoids unsupported production claims.

Project Timeline

Indicative Delivery Timeline



Timeline is indicative for portfolio planning. Actual sequencing depends on data readiness and stakeholder review cadence.

Figure 4. Indicative timeline for BI portfolio delivery.

Phase gate	Exit condition
Discovery complete	Scope, stakeholders, source data and KPI candidates are signed off.
Data build complete	SQL marts reconcile key totals and document known exclusions.
Analysis complete	Python findings are evidence-backed and tied to business objectives.
Dashboard complete	Power BI pages satisfy acceptance criteria and executive review questions.

Appendix

A. Source of Truth Summary

Fact	Approved value
Company	NovaCart
Business model	B2C Multi-Seller Marketplace
Customers	185,000
Active sellers	520
Products	24,500
Warehouses	8
Annual orders	550,000
Revenue FY2025	₹185 Crore
Profit FY2025	₹16.5 Crore

B. Requirement Traceability Notes

- Functional requirements define what the BI solution must provide.
- Non-functional requirements define usability, security, performance and maintenance expectations.
- Data requirements define the minimum fields and grains needed for SQL and Python work.
- Business rules protect consistency with the Company Profile and future portfolio documents.
- Dashboard requirements define how executive and functional stakeholders will consume the analysis.

C. Downstream Artifact Dependency

Artifact	BRD dependency
KPI Dictionary	Success metrics, business rules and reporting requirements
SQL	Functional, data and business-rule requirements
Python	Analytics requirements and accepted business questions
Machine Learning	Feature-ready data requirements and ML exploration boundaries
Power BI	Dashboard requirements, success metrics and reporting cadence